

Day 1 — The Hook: What Does “Indian Knowledge” Mean?

Module: What is IKS? (Middle) · **Time:** 45 min

Learning objective

By the end of this lesson, students will give an open-ended definition of “Indian Knowledge Systems” that mentions at least three distinct domains (e.g. mathematics, medicine, philosophy).

Materials

- Whiteboard
- Sticky notes (3 per student)
- Wall poster: “Indian contributions you’ve encountered THIS WEEK” (to be filled together)

Vocabulary

- *vidyā* — “knowledge, learning”
- *śāstra* — “treatise; systematic body of knowledge”
- *jñāna* — “knowledge as understanding” (different shade than *vidyā*)

Flow

Warm-up (5 min)

Question on board: “*What’s one Indian-origin idea you’ve encountered this week, outside this class?*”

Students write on sticky notes. Post on the wall. Read aloud the variety: yoga, zero (in math), turmeric (kitchen), an Indian-named star (Rigel? actually Arabic — but the question is fertile), namaste at the start of a class, an Ayurvedic ingredient on a shampoo bottle.

Core (20 min) — A working definition

1. **The question:** “*If someone said to you ‘tell me about Indian knowledge,’ what would you say?*”

- Take 3–4 student answers. Don’t correct. Note variety.

2. **Build a working definition** on the board:

Indian Knowledge Systems (IKS) = the intellectual traditions developed on the Indian subcontinent over ~3000 years — including science, mathematics, astronomy, linguistics, philosophy, medicine, music, architecture, and more.

3. **Three things IKS is NOT:**

- **NOT only religion.** (Some IKS is religious; much is not.)

- **NOT magical thinking dressed up as science.** (Real claims, real methods.)
 - **NOT a competitor to modern science.** (Both can be true at the same time.)
4. **Three things IKS IS:**
- A real body of intellectual work with specific authors, dates, and texts.
 - Sometimes overlapping with, sometimes distinct from, religious tradition.
 - Worth studying as history of human inquiry.
5. **One specific example to anchor today:**
- The Indian invention of **zero** as a number (not just placeholder) is documented in Bakhshali manuscript (~7th c. CE), and used in Bhāskara II's *Līlāvatī* (12th c. CE). It travelled via Arab mathematicians to Europe by ~12th c.
 - That's a real, dated, sourceable contribution.

Activity (15 min) — Inventory

Each student writes on 3 sticky notes: 1. One thing they **ALREADY KNEW** about IKS (before today). 2. One thing they **WANT TO LEARN** about IKS in this module. 3. One thing they're **SKEPTICAL** of (any claim about IKS they doubt).

Post on the wall. Cluster by theme during break.

Wrap-up (5 min)

- Note out loud: “By Day 10 you’ll each give a 5-minute presentation on **ONE** specific Indian contribution. Start thinking about which strand interests you most.”
- Preview Day 2: “*Tomorrow — the Vedas. Not the religious part you might have heard about. The intellectual part.*”

Student-facing content

What is IKS?

Indian Knowledge Systems = the intellectual traditions of the Indian subcontinent (~3000 years).

Includes: mathematics, astronomy, medicine, linguistics, philosophy, music, architecture.

One example: **the number zero**. First documented as a number (not just a placeholder) in the **Bakhshali manuscript** (~7th century CE). Used in Bhāskara II's *Līlāvatī* (1150 CE). Reached Europe by the 12th century via Arab mathematicians.

Homework

- Read the project brief ([assessments/project-brief.md](#)). Think about which strand of IKS you might pick.

Differentiation

- **Tier up:** Find **ONE** thing you previously believed about IKS that turned out to be wrong (or oversold). Bring tomorrow.

- **Tier down:** Just remember the working definition: *intellectual traditions of the Indian subcontinent*.

Teacher notes

- The “what would you say?” prompt often reveals real diversity — some students will say “religion,” some “yoga,” some “math.” All are partly right.
- The “NOT only religion” framing is important early. Many students conflate IKS with Hinduism. They overlap; they are not identical.

Citation(s) used

- vidya-karana RAG cache day-01-intro.json for the general framing.
- Plofker (2009) for the zero/Bakhshali reference.